Understanding and Tuning Catalytic Performance of Alumina via Oxygen Vacancies

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Oxygen vacancies play a crucial role in catalytic reactions such as CO_2 hydrogenation, the water-gas shift reaction, and NO reduction.[1,2] However, the mechanisms behind their formation and participation in these reactions are not yet fully understood. Additionally, an optimal concentration of oxygen vacancies has been identified for various applications, highlighting the importance of tuning catalytic materials to achieve ideal vacancy formation under reaction conditions.

Alumina-based materials are widely used as catalyst supports in industrial dehydrogenation processes. Recent findings indicate that different phases of Al_2O_3 exhibit intrinsic dehydrogenation and cracking activity, even without the addition of noble metals, with the support material playing a crucial role in coke deposition and catalyst deactivation.[3] This study aims to establish a relationship between oxygen vacancy formation energies and activity and stability of Al_2O_3 catalysts in the propane dehydrogenation. Density functional theory (DFT) calculations were employed to identify oxygen vacancy formation energies at the Al_2O_3 (001) surface and their role in the reaction mechanisms of propane dehydrogenation and coke formation. We further examine the influence of transition metal dopants on oxygen vacancy formation, activity, and stability.

Our computations reveal that under industrially relevant reaction conditions, isolated oxygen vacancies can be formed. Hydrocarbons exhibited significantly stronger adsorption at or near these vacancies. Furthermore, the vacancy acts as the acceptor of the first hydrogen abstracted during the dehydrogenation mechanism. These results are supported by catalytic measurements on Al_2O_3 powders, which demonstrated that the catalytic activity and selectivity of Al_2O_3 increased in the presence of hydrogen and decreased in the presence of water. We further propose the first step-by-step chain

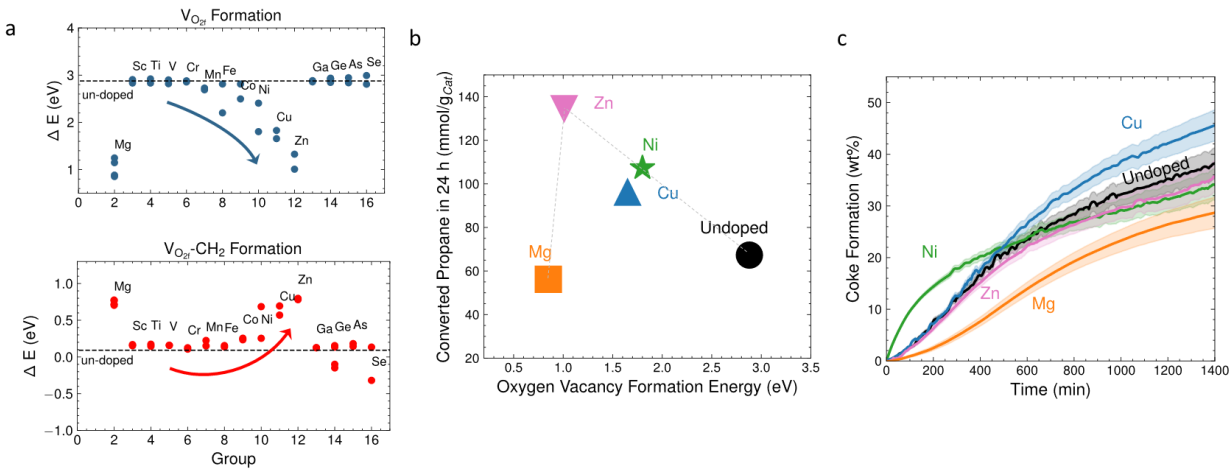

Figure 2: a) Calculated formation energy of oxygen vacancy and coke precursor for doped Al_2O_3; b) Relationship between oxygen formation energy and measured catalytic activity; c) *In-situ* coke formation measurements of doped Al_2O_3.

growth mechanism toward the formation of coke and catalyst deactivation, where a deep-dehydrogenated C3 species is found to adsorb strongly in those vacancies and enable chain growth with additional propylene molecules, leading finally to coke formation. To mitigate the formation and stabilization of these coke precursors, we aimed to predict doping elements that could influence catalytic performance and reduce deactivation (Figure 1a). Promising dopants (Mg, Ni, Cu, Zn) were then incorporated into alumina using a sol-gel synthesis method and tested in a fixed-bed reactor and thermogravimetric suspension balance.[4] Analytics confirmed the successful formation of a γ-Al_2O_3 structure with well dispersion of the doping elements. The materials doped with Zn, Cu and Ni display significantly enhanced activity (Figure 1b), while maintaining the high stability of Al_2O_3, whereas Mg suppresses both, coking and the PDH reaction (Figure 1c). Connecting these results with our DFT calculations, we firmly establish a relationship between the oxygen vacancy formation energy and PDH catalytic activity (Figure 1b). Our proposed coking mechanism established in this study provides fundamental insight into the elementary steps of this degradation process and enables rational development of coking-resistant catalysts.

References

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